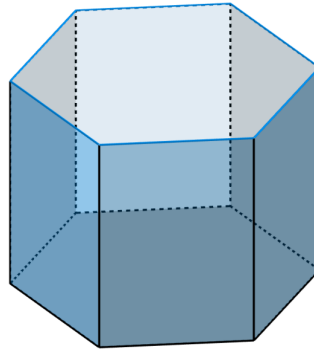




9.3.5 Wrap-Up: Ticket Out the Door

1. A hexagonal prism is shown.



- a. Draw the cross section that results from a plane that is parallel to the base.



- b. Draw the cross section that results from a plane that is perpendicular to the base.



Unit 9, Lesson 4: Solids of Rotation



Benchmark

MA.912.GR.4.2 Identify three-dimensional objects generated by rotations of two-dimensional figures.



Learning Target

- ☐ I can identify three-dimensional objects generated by rotations of two-dimensional figures.



9.4.1 Warm-Up: Notice and Wonder

A Rubik's Cube and a Slinky are shown.



1. What do you notice?
2. What do you wonder?
3. Sketch the horizontal cross section of the Rubik's cube.
4. Sketch the horizontal cross section of the Slinky.



9.4.2 Activity: Rotating Pencils

Work with your partner to complete the following.

1. Place a sticky note on the side of the pencil. If necessary, use tape to secure the sticky note.
2. Make a prediction about which three-dimensional shape the sticky note would form when the pencil is rotated 360° .

Complete the table.

	Prediction	Reasoning
Sticky Note		

3. Using your pencil as a vertical axis of rotation, rotate the pencil 360° .
4. What three-dimensional figure did the sticky note seem to create?

5. Work with your partner to predict what three-dimensional shape will be formed when each shape is rotated 360° .

Shape	Prediction	Reasoning
Rectangle		
Triangle		
Semicircle		

6. Using your pencil as a vertical axis of rotation, rotate the pencil 360° . Then, sketch or describe the resulting three-dimensional shape.

Shape	Three-Dimensional Shape
Rectangle	
Triangle	
Semicircle	

7. Discuss with your partner whether the three-dimensional shape would be different if the pencil were held horizontally instead of vertically. Summarize your discussion.

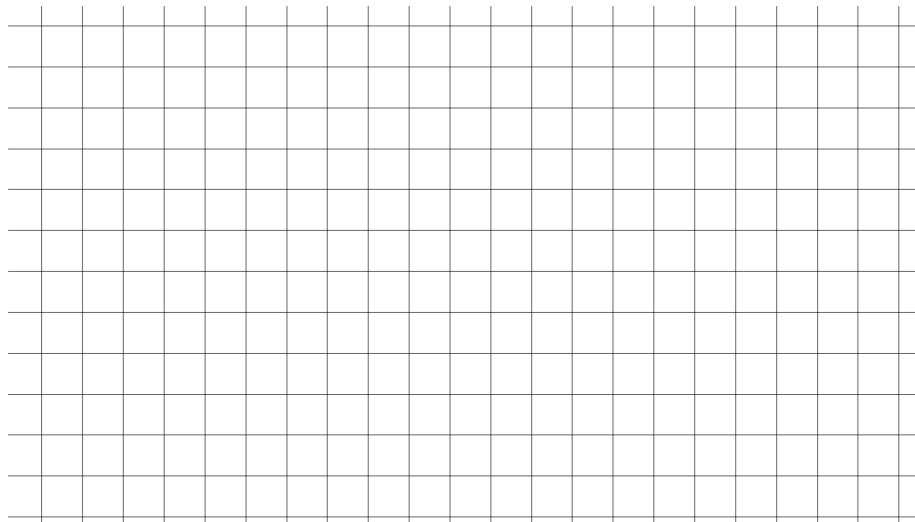
8. Draw any four-sided, two-dimensional shape on a 3x5 card. Make it as unique as possible. Cut out your shape. Draw the outline of your shape below.

9. Sketch what you think the three-dimensional figure of your four-sided two-dimensional shape will look like if rotated around each axis.

Horizontal Axis	Vertical Axis

10. Review your partner's shape. Then, complete the following.

- Draw your partner's shape on the grid below.
- Draw a horizontal axis of rotation.
- Draw the reflection of your partner's shape.



11. Discuss with your partner how drawing a reflection could help envision the three-dimensional figure. Summarize your discussion.

12. Sketch what you think the three-dimensional figure of your partner's four-sided, two-dimensional shape will look like if rotated around each axis.

Horizontal Axis	Vertical Axis

13. Compare your sketches with your partner. Discuss any differences and make any adjustments.

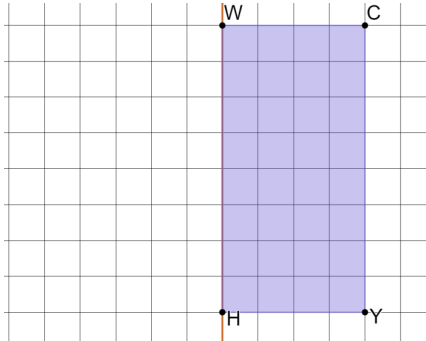
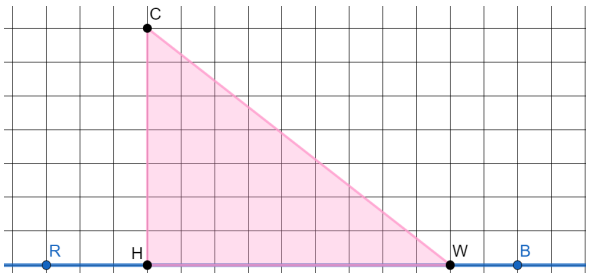


9.4.3 Guided Instruction: Rotations With Dimensions

1. Draw a two-dimensional figure and an axis of rotation that would result in the real-world object when the two-dimensional figure is rotated around the axis of rotation.

Image	Two-Dimensional Figure
A set of revolving doors with a radius of 3 feet. 	
An ice cream cone with a diameter of 4.5 centimeters (cm) and a height of 11 cm. 	

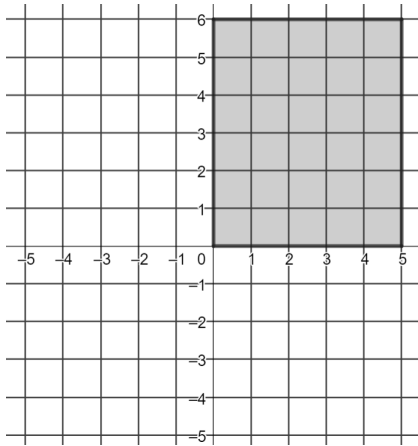
2. Sketch the three-dimensional shape that is created. Include any dimensions in units.

Two-Dimensional Shape	Three-Dimensional Shape
$WCYH$ is rotated around $\overleftrightarrow{WH}$. 	
$\triangle CWH$ is rotated around $\overleftrightarrow{RB}$. 	

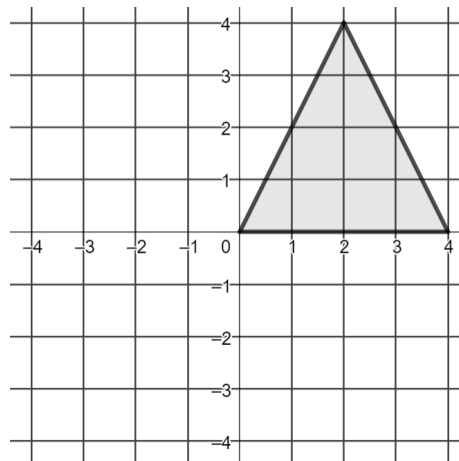


9.4.4 Guided Practice: Rotations

1. Describe and sketch the three-dimensional shape that is created when the rectangle is rotated around the y -axis. Include any known dimensions.



2. Describe and sketch the three-dimensional shape that is created when the isosceles triangle is rotated around the y -axis. Include any known dimensions.



9.4.5 Wrap-Up: Ticket Out the Door

1. Describe and sketch the three-dimensional shape that is created when the semicircle is rotated around its diameter. Include any known dimensions.

